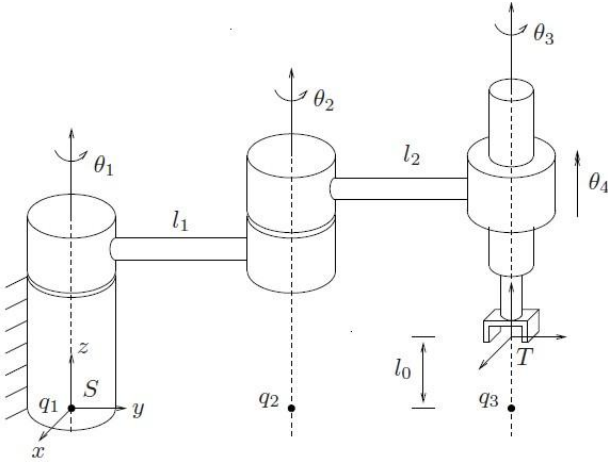


Tutorial assignment May 13, 2026

Consider a 4DOF manipulator shown in reference configuration in the figure below.



In the reference configuration

- the links of length l_1 and l_2 are both parallel to the spatial y axis,
- the tool frame axes are parallel to the corresponding spatial frame axes, and positive in the same direction, and
- the tool frame origin is a distance l_0 higher (in the spatial z direction) than the spatial frame origin.

$l_0 = 0.3$ m, $l_1 = 1.2$ m and $l_3 = 0.8$.

The manipulator now is moved with joint rotations

$$\theta = \begin{bmatrix} 15 \\ 54 \\ -32 \\ 0.05 \end{bmatrix},$$

angles **in degrees** and displacements in m. At the instant this new configuration is reached, the joint velocities are

$$\dot{\theta} = \begin{bmatrix} 0.4 \\ -0.2 \\ 0.28 \\ 0.1 \end{bmatrix},$$

in rad/s or m/s.

Let q_T be the origin of the tool frame.

1. Calculate the instantaneous velocity vector $v_{q_T} \in \mathbb{R}^{3 \times 1}$ in spatial frame components.
2. Calculate the body velocity of the end effector frame $V_{st}^b \in \mathbb{R}^{6 \times 1}$.
3. With the manipulator in the current configuration, calculate the joint torques (six different answers) for the following six separate cases F_1 to F_6 of end effector wrench loading (in N and Nm), specified

below in end effector body frame components, and comment on whether the answers make sense. Ignore any possible effect of accelerations.

4. Calculate the corresponding six wrenches expressed in spatial frame components, to F_1 to F_6 , with the manipulator in the current configuration.
5. Check whether the calculation of the six sets of joint torques based on the six wrenches expressed in spatial frame components renders the same results as those calculated in part 3.

$$F_1 = \begin{bmatrix} 100 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad F_2 = \begin{bmatrix} 0 \\ 100 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad F_3 = \begin{bmatrix} 0 \\ 0 \\ 100 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad F_4 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 100 \\ 0 \\ 0 \end{bmatrix} \quad F_5 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 100 \\ 0 \end{bmatrix} \quad F_6 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 100 \end{bmatrix}$$